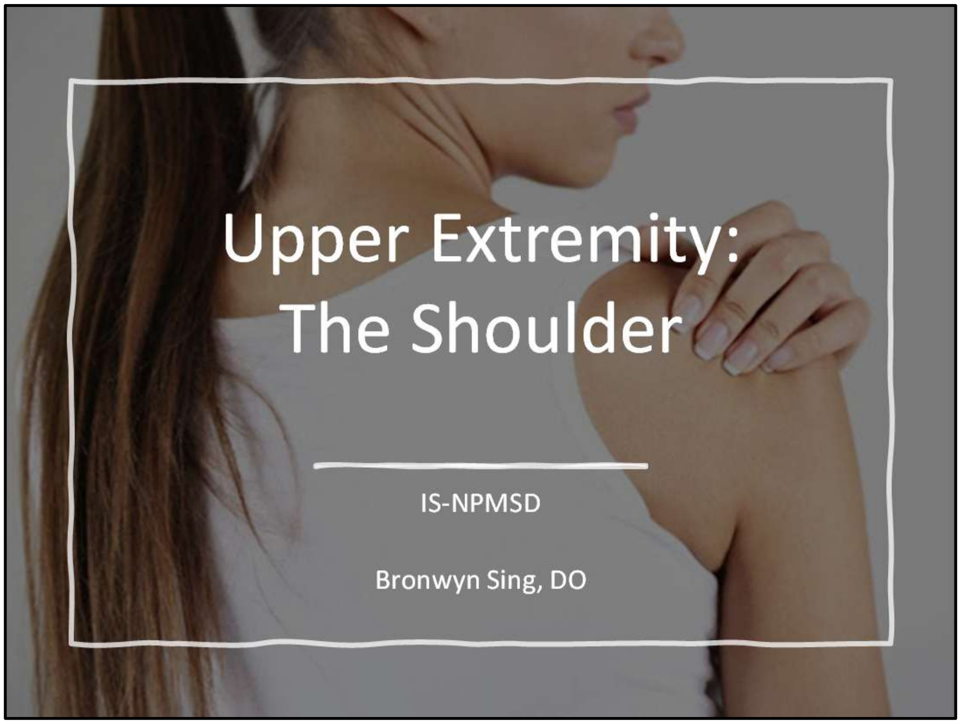

Upper Extremity: The Shoulder

IS-NPMSD

Bronwyn Sing, DO

Lecture Objectives

- Recognize common shoulder presentations in primary care and MSK settings.
- Describe pathophysiology and anatomy of key shoulder pathologies.
- Perform focused shoulder exam and recognize ultrasound opportunity
- Differentiate pathologies based on history, physical exam, and imaging.
- Outline initial non-surgical management and referral indications.

Approach to Shoulder Pain

1. History: Onset, trauma, activity/ADLs, pain characteristics, instability, weakness.
2. Physical Exam: Inspection, Palpation, ROM, Strength testing, +/- neurovascular, special tests.
3. Consider Ultrasound: Assess rotator cuff, biceps tendon, AC joint, effusion.
4. Key Diagnostic Decision: Painful vs Weak shoulder, Traumatic vs Atraumatic.

Functional Shoulder Review

- Shoulder connected to axial spine via clavicle and scapula.
- Scapulothoracic motion can impact shoulder function.
- Glenohumeral joint: shallow, maximal mobility but also least stable. Labrum provides depth and limits anterior translation at inferior portion
- SITS muscles: dynamic stabilizers.

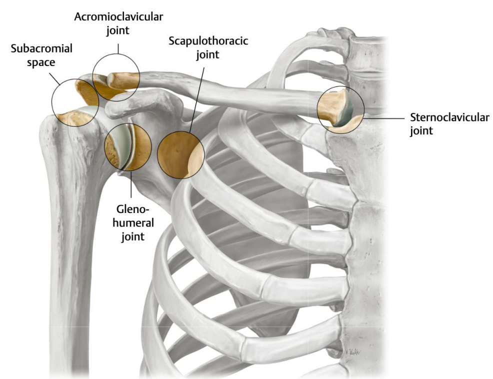

Fig. 25.7 **Joints of the shoulder: Overview**
Right shoulder, anterior view.
Source: Joints of the Shoulder. In: Gilroy A, MacPherson B, Wikenheiser J, Schünke M, Schulte E, Schumacher U, Voll M, Wesker K, ed. Atlas of Anatomy. 5th Edition. New York: Thieme; 2025. doi:10.1055/b000001037

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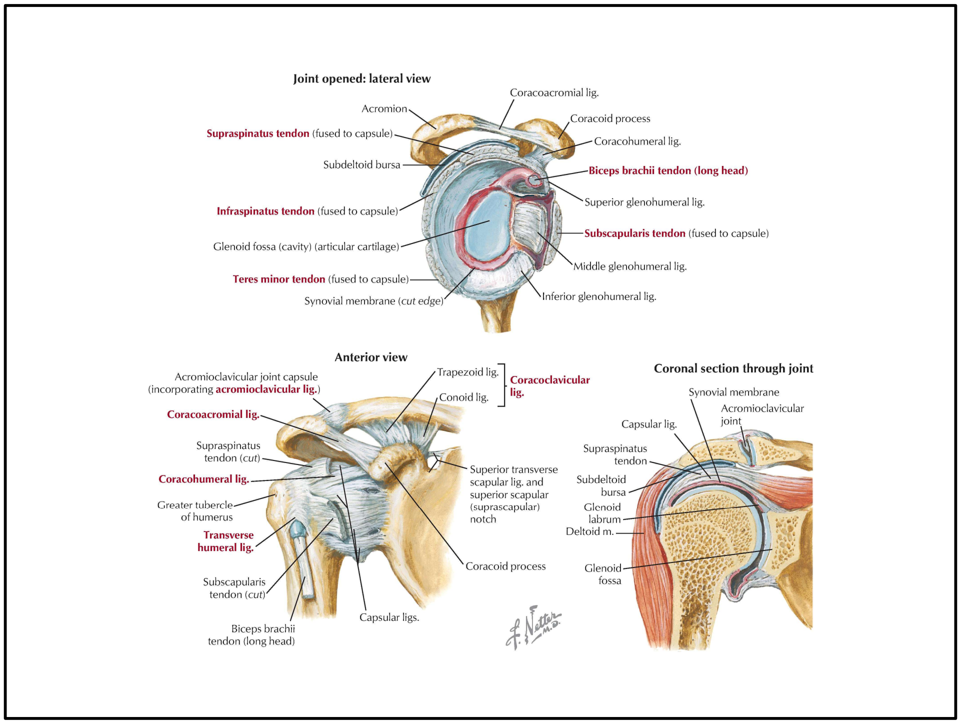
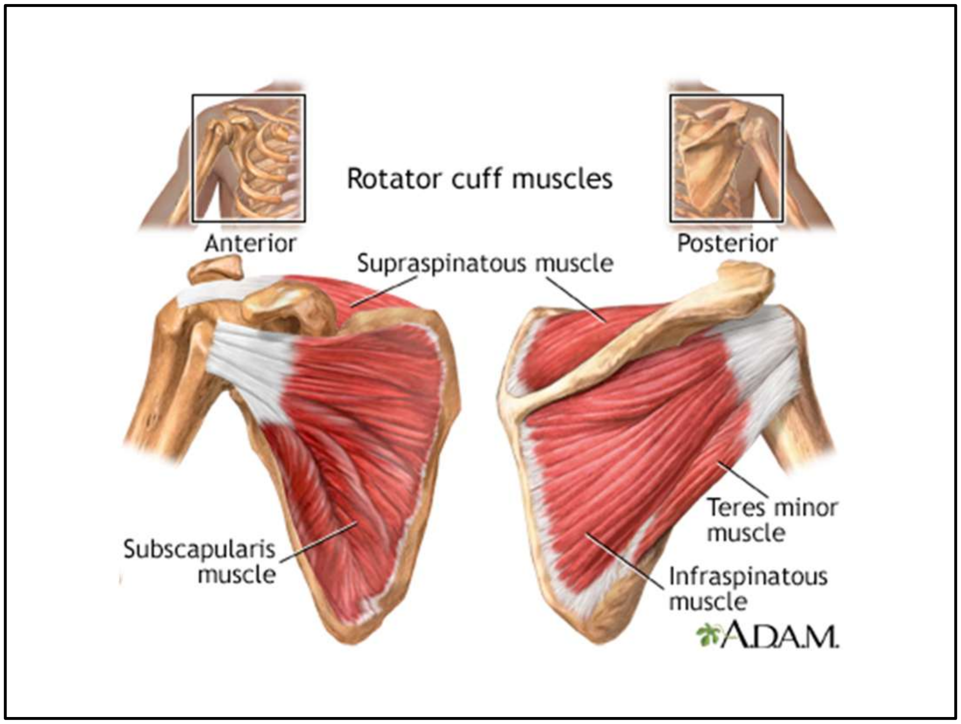
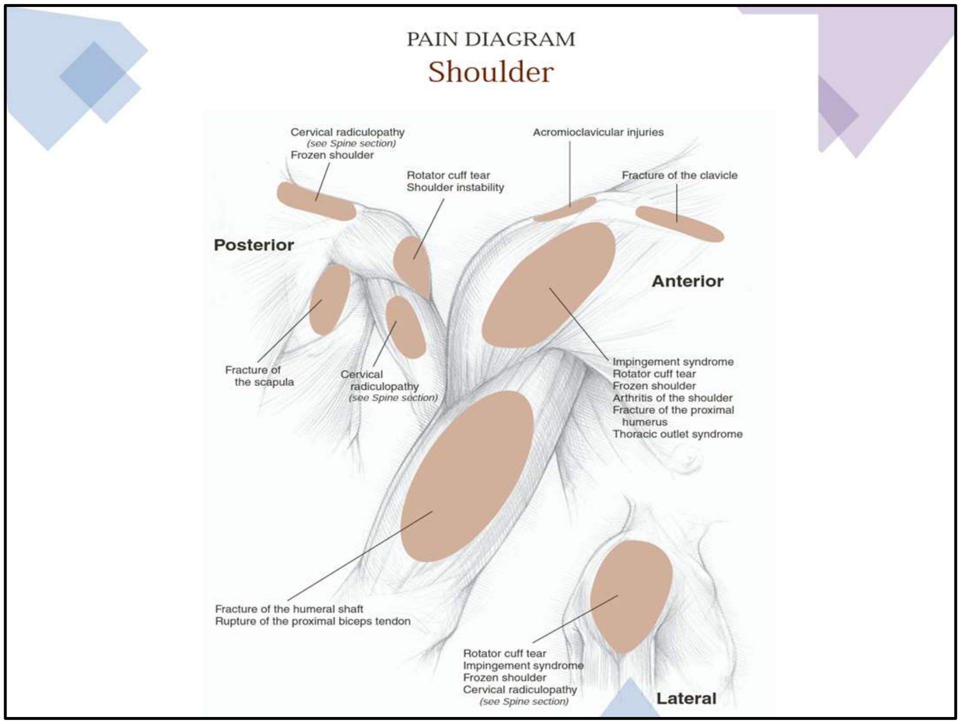


Glenohumeral joint: shallow, maximal mobility but also least stable. Labrum provides depth and limits anterior translation at inferior portion

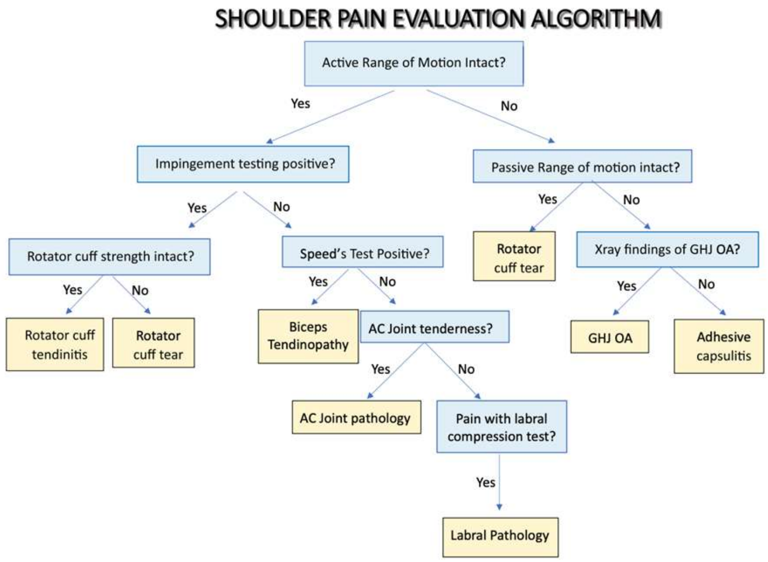


SITS muscles: dynamic stabilizers.



Finger point test:
Lateral pain- GH
Top of shoulder- AC joint
Posterior Shoulder- consider Cervical cause

Figure 1. Shoulder pain evaluation algorithm.



Note: GHJ, Glenohumeral Joint; OA, Osteoarthritis; AC, Acromioclavicula.

Inspection, Palpation, Range of Motion, Strength

Ultrasound in Shoulder Assessment

- Dynamic and point-of-care assessment tool for shoulder evaluation.
- Common views: Supraspinatus (rotator cuff), Biceps tendon, Subacromial bursa, AC joint.
- Key findings:
  - **Hypoechoic gap → Rotator cuff tear**
  - Fluid in bursa → Bursitis / Impingement
  - Widened AC joint → Separation
  - Effusion posterior recess → Glenohumeral effusion

This does not replace a solid physical exam, you need to know the pain generator. But it can help you decide next steps earlier. i.e) steroid injections, Xray, MRI.

Frieda

A 46 yr old female presents to your clinic with R shoulder pain x 2months. She denies trauma and reports that the pain has slowly worsened.

She admits to localized pain over her deltoid and reports that she cannot move her arm overhead.

Frieda admits to starting a new job at a warehouse where she is reaching for packages over head. She is right handed.

What do you want to know about Frieda in terms of history??

What is in your differential

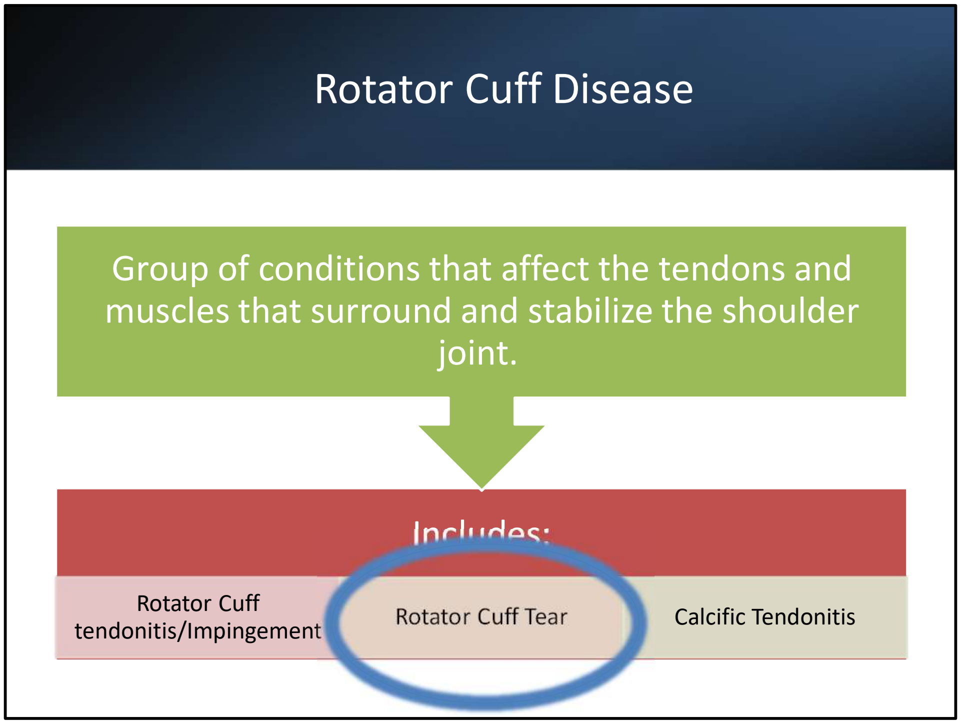
Vital Signs are stable, her BP is mildly elevated at 134/78.

DDX for shoulder pain

- **Non traumatic:**
  - Rotator cuff disease
    - Overuse, degenerative
  - Adhesive capsulitis
  - GH arthritis
  - Biceps tendonitis
  - Cervical radiculopathy

Traumatic:

- Rotator Cuff Disease
  - Trauma
- AC joint separation
- Fracture
- Shoulder dislocation/subluxation



These all have a common constellation of symptoms:

Pain with overhead activity, location is usually lateral shoulder. Night pain/discomfort.

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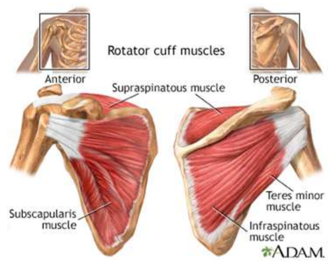
- RTC tendonitis- overuse- gradually worsens, pain with AROM but intact, can do passive ROM, but no weakness
- RTC TEAR- trauma or degenerative (will have hx of shoulder pain/issues) ROM not intact, Passive ROM intact, WEAKNESS
- Calfcific Tendonitis- rapid onset pain, wakes them from sleep, pain is so high, they can even present to the ER.

Rotator Cuff Tear

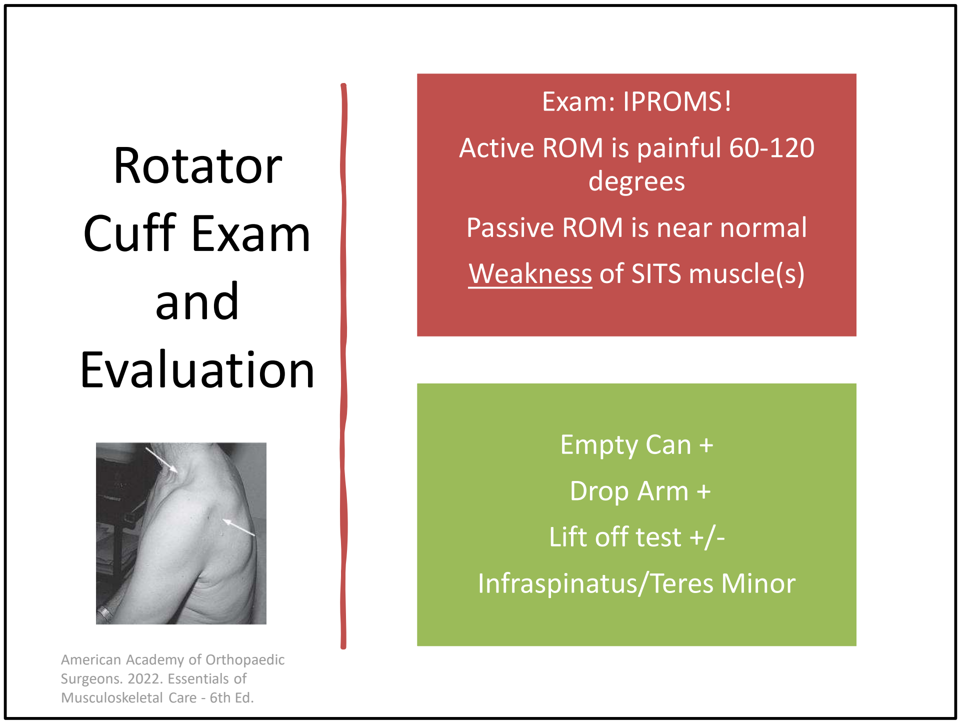
<u>Epidemiology</u>: Consider age group. Most common cause of shoulder pain in adults >40. Age 65+ high risk tears, may be asymptomatic

<u>Pathophysiology</u>: **Degenerative** or **traumatic** tear of supraspinatus/infraspinatus.

<u>History</u>: Overhead activity. Symptoms of lateral shoulder pain, night pain, weakness lifting arm, painful arc 60–120°.



Insidious onset, worse with overhead activities and when lifting objects away from body.
May or not identify a traumatic injury.
Patient will often point to deltoid (lateral) region.
Pain can be minimal, aching and dull. Night pain and difficulty sleeping on the affected side are characteristic of a rotator cuff tear.
Symptoms include WEAKNESS, catching pain when raising the arm overhead (60-120 degrees)- PAINFUL ARC



Inspection: Back of the shoulder may appear sunken with chronic presentation.

Indicates atrophy of infraspinatus muscle following long-standing tear.

Palpation: May have tenderness over the greater tuberosity

ROM:

Active range of motion painful past 60-120degrees ABDuction (painful arc)

Passive range of motion is near normal

Weakness is appreciated with one or more of rotator cuff muscle testing usually supraspinatus. Mild weakness-4/5, partial tear. Full weakness is a full tear.

TEST SITS:

Empty Can- supraspinatus

Drop Arm- Supraspinatus full tear?

Lift off test- Subscapularis

Infraspinatus/ Teres minor Resistence- tests those two muscles together.

Rotator Cuff: Imaging

- X rays may show a high riding humeral head
- MRI: Provides additional information on the status of the muscle and on the size of full-thickness and some partial-thickness tears.

Coronal oblique MRI of the shoulder demonstrates a rotator cuff tear (arrows). (UptoDate 2022)

Space between humeral head and acromion is estimated to be 7-14mm, if <7mm on xrays can suspect large rotator cuff tear.
MRI is accurate 93-100% for full thickness tears, less so for partial tears.
MRI advantage is the quality of rotator cuff, size of tear, involvement of biceps tendon, the presence of partial thickness tear.

Ultrasound for decision making, fast track referrals.
MRI for surgical planning

Figure 2. Normal supraspinatus tendon

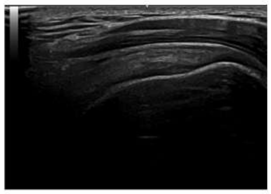
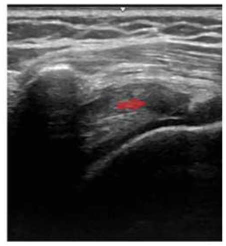
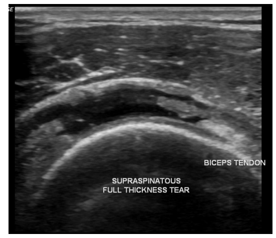


Figure 6. Supraspinatus partial tear.





Ultrasound shoulder findings

- Ultrasound: Hypoechoic or anechoic gap, tendon discontinuity.

https://www.jabfm.org/content/jabfp/37/6/1156.full.pdf
Radiopedia RCT

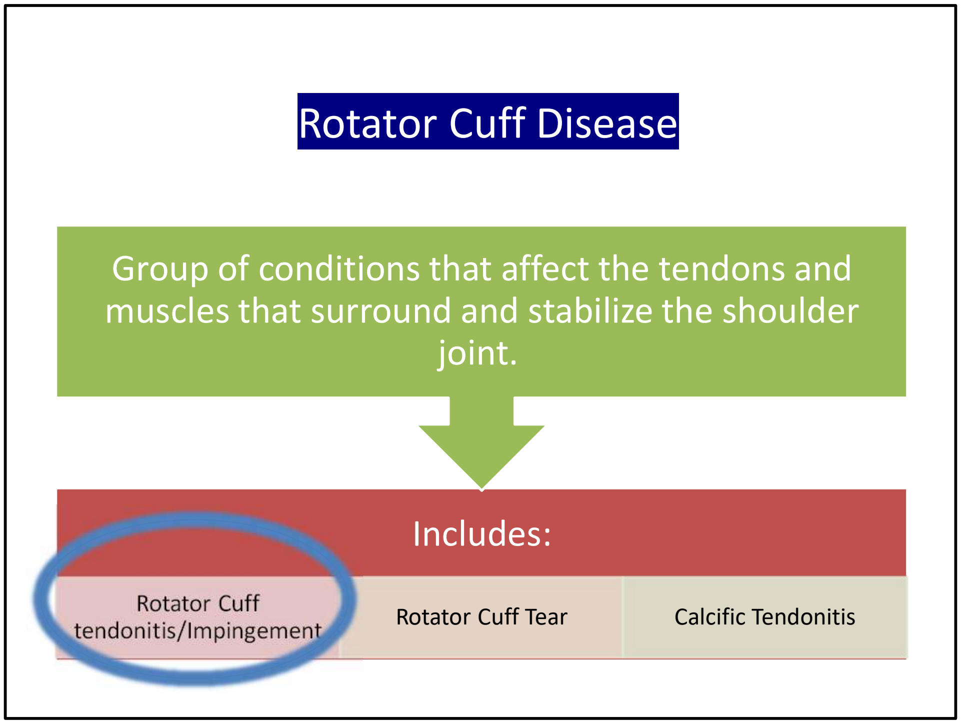
Rotator Cuff Management

- Management:

  - Who should get surgery? Consider function.
  - Full thickness tears get MRI and referral to Ortho!

  - NON-Operative: PT, NSAIDs, OMM, steroid injection
  - OPERATIVE: MRI and surgery if acute full tear

Use injections judiciously. May decrease inflammation and provide short term pain relief but also weakens the tendon. repeated injections may ultimately accelerate propagation of the rotator cuff tear
After 8 weeks, tendon will retract and will not be able to surgical repair



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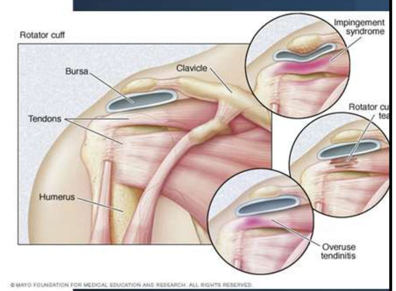
Rotator cuff Tendonitis/Impingement

- **History**: Repetitive overhead activity → compression of supraspinatus tendon/bursa under acromion leads to inflammation
- **Symptoms**: Gradual anterior/lateral pain, worse overhead, night pain when lying on that side
- **Exam**: IPROMS
  - Painful arc but AROM and PROM intact
  - Strength: <u>intact</u>
  - Neer +, Hawkins +



Subacromial impingement is the MOST COMMON cause of shoulder pain • Accounts for 44-65% of shoulder disorders

Tenderness to palpation over the greater tuberosity and subacromial bursa

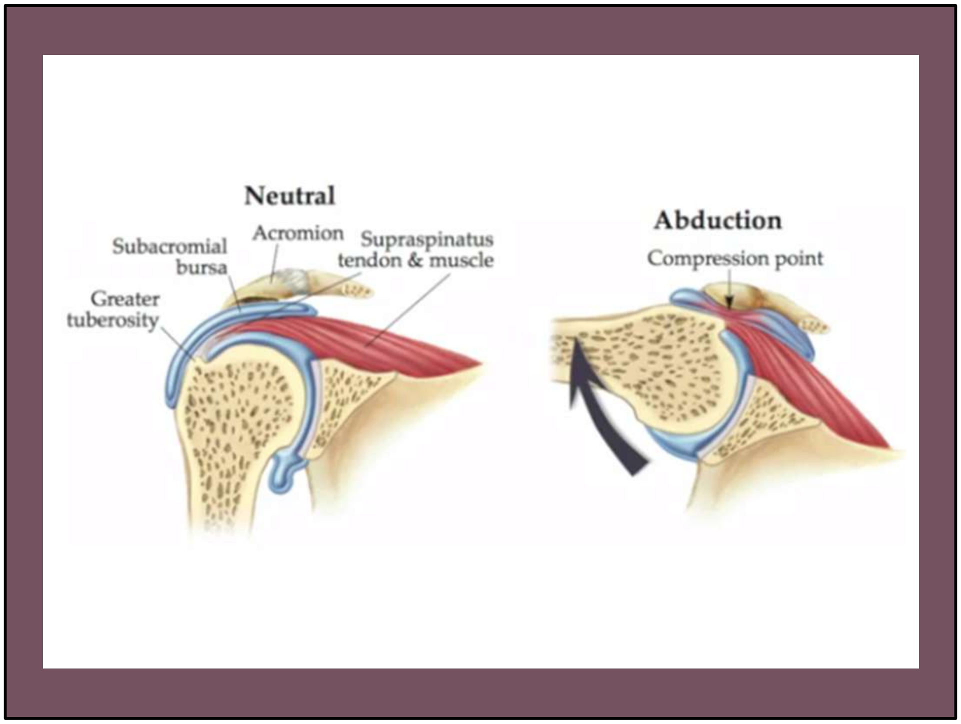
Crepitus with shoulder motion

Pain worse between 90 to 120 of

abduction when lowering the arm

Positive Neer and Hawkins sign, Empty Can sign with pain only

Strength is usually normal (rotator cuff strength intact)



Recognize how many structures are under the AC joint and any pathology of those can cause impingement syndrome
The rotator cuff, primarily the supraspinatus tendon is pulled repetitively under the coracromial arch with overhead movement.
Bursa and tendons can become inflamed.
Impingement when the space between the acromion and the rotator cuff narrows

Impingement Syndrome cont’d

ion of SA-SD bursa. The tip of a 25 gauge needl
s (arrowhead). A combination of corticosteroid
ıatus tendon; g = greater humeral tuberosity.

- Ultrasound: Subacromial bursal fluid
- Diagnostic Tests: X rays usually normal

Australas J Ultrasound Med
. 2015 Dec 31;13(2):11–15. doi: [10.1002/j.2205-0140.2010.tb00151.x](10.1002/j.2205-0140.2010.tb00151.x)

Diagnostic Tests: X rays usually normal.

Subacromial impingement

- Treatment: Activity modification, NSAIDs, PT, Steroid injection
- OMM to improve scapular mechanics, upper cross syndrome and posture
- Strengthen muscles to anchor scapula

Diagnostic injection can be done

- Local anesthetic is injected into subacromial space and provides complete pain relief
- Increased strength on “empty can” test

Calcific tendonitis

- Patients often present with pain at the top or lateral aspect of the shoulder or both, often with radiation towards the insertion of the deltoid.
- The onset of **pain is rapid and unassociated with trauma. Pain level is high!!**
- Most patients report increased pain at night and an inability to lay on the affected shoulder. Daily activities involving overhead motions are affected.
- Physical Exam:
  - Acute episode of pain due to calcific tendinopathy often refuse to move their shoulder, holding their arm close to their bodies in internal rotation.
  - \+ Neers and + Hawkins sign

Plain radiograph of calcific tendinopathy of the shoulder in the resting (chronic) phase

This plain radiograph of the shoulder shows calcific deposits (arrow) in the region of the supraspinatus tendon. The deposit is dense and homogeneous with well-defined limits suggesting the tendinopathy is in the resting phase.

Courtesy of Tore Prestgaard, MD.

UpToDate

Mechanically, calcium deposits can reduce the space between the rotator cuff and the acromion causing subacromial soft tissues to become impinged. The deposits and reduced space can also impair the normal function of the rotator cuff. Inflammation during resorption of calcific deposits can produce pain.

Studies indicate that the presence of calcification is about 50% symptomatic/50% asymptomatic.

Calcific tendonitis

- X rays will help diagnosis with visualization of calcifications along shoulder tendons.
  - Clear margins: Chronic
  - Fluffy/Poorly defined margins: Resorptive Phase

- Formation of calcifications are not well understood, no relation with overuse
  - May have a connection with DM or thyroid disease
- Management is similar to impingement

Plain radiograph of calcific tendinopathy of the shoulder in the resorptive phase

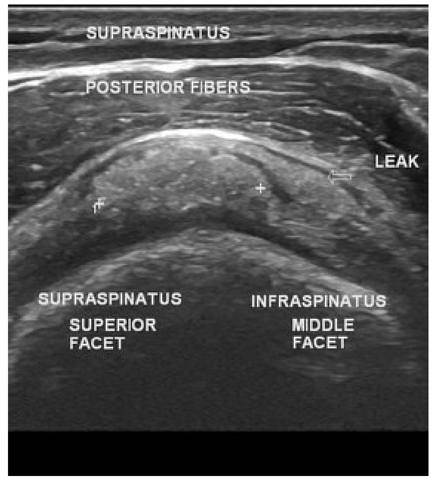
The plain radiograph above shows calcific deposits (arrows) in the shoulder. The deposits have a fluffy appearance with poorly defined limits, suggesting that the patient's calcific tendinopathy is in the resorptive phase.

Courtesy of Tore Prestgaard, MD. UpToDate

X rays to order: Anteroposterior (AP), outlet, and internal and external rotation views should be obtained. (which views to order is not tested)

Calcific Tendonitis

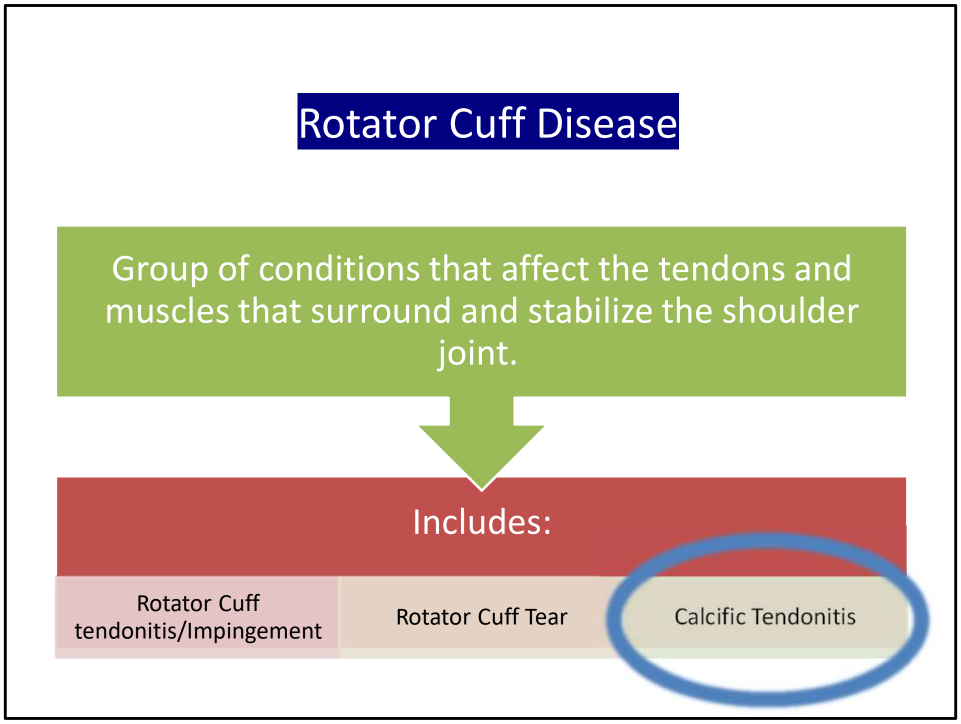
- Ultrasound findings:



Frieda

- Inspection: WNL
- Palpation: TTP over deltoid region
- ROM: limited AROM. Flexion to 90 and Abduction to 90 due to pain, passive ROM limited due to pain
- Empty can test elicits pain but no weakness.
- Lift off, strength for infraspintus/teres minor 5/5. All other rotator cuff mm intact.
- Positive Neers and Hawkin's sign, strength remained intact.
- What do you suspect her diagnosis?

Treated with NSAIDs, gentle ROM, follow up for OMM.



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To Review!

Feature	Rotator Cuff Tear	Calcific Tendonitis	Shoulder Impingement Syndrome
Etiology	Degenerative, traumatic	Calcium deposition (etiology unclear)	Mechanical compression
Typical Symptoms	Weakness, night pain, reduced ROM	Intense pain, tenderness, restricted ROM	Pain with overhead activities, night pain
Diagnosis	MRI or ultrasound	X-ray or ultrasound	Clinical, MRI/US if tendinopathy suspected
Management	Conservative, surgery for full-thickness tears	Conservative, needling, shockwave therapy	Conservative, surgery for refractory cases

Chart taken from chatgpt inquiry: OpenAI. (2024). *ChatGPT* (https://chat.openai.com/chat. Compare rotator cuff tear, calcific tendonitis, shoulder impingement syndrome.

Baby shark

SHOULDER AND UPPER ARM

biceps tendonitis

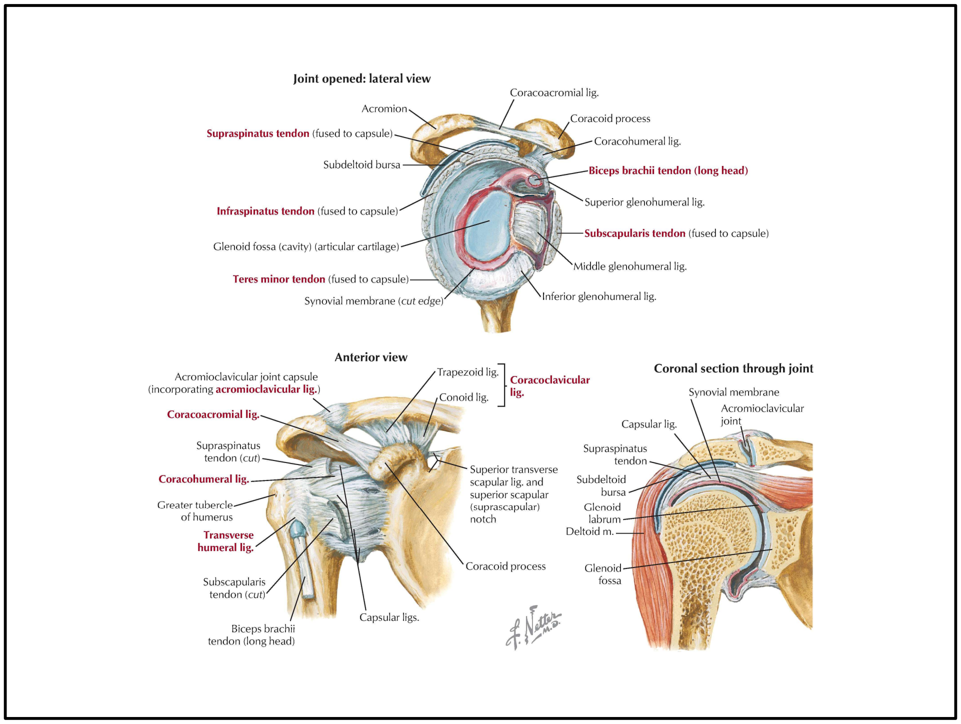
- Pain with arm flexion. Very common in conjunction with subacromial bursitis
- Can be proximal or distal
- Exam:
- Pain in bicipital groove, Speed's/Yergason's +.

- Compensation by using only lower arm movements
- Treatment: Ice, heat, exercises, physical therapy
  - injections are very effective

Speed's Test Video

Yergason's Test Video

Long head of the biceps inserts



Note long head of the biceps and its relation to joint capsule and labrum

biceps tendon
rupture

SHOULDER AND UPPER ARM

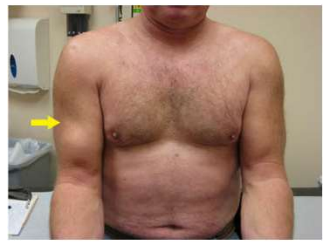
biceps tendon rupture

- Rupture of the proximal long head of the biceps
- Older adults with history of rotator cuff disease
- Uncommon in young adults but may occur in athletic individuals involved in weight lifting or throwing sports
  - Inflammation in the shoulder capsule can affect the tendon

Long head of the biceps ascends in the inter tubercular groove and is intra-articular for its proximal 3cm.

biceps tendon rupture clinical presentation

- Sudden pain in the upper arm, often accompanied by an audible snap
- Patients may notice a bulge in the lower arm
- Pain in the acute phase is often mild



What is popeye's sign?

Not this!

biceps tendon rupture physical exam

- Inspection: + Ecchymosis (acute). Bulge in the lower arm resulting from the muscle belly of the biceps retracting into the lower arm
  - ROM/Strength: Have the pt contract the biceps against resistance- accentuated bulge
- Defect can be palpated proximal end
- Gentle pressure over the bicipital groove of the humerus with the arm in 10 of internal rotation will elicit acute pain

SHOULDER AND UPPER ARM

biceps tendon rupture diagnosis

- Diagnostic tests
  - AP and axillary radiographs to rule out fracture
  - Ultrasound: Ultrasound: Effusion in tendon sheath or absent long head tendon.
  - MRI can rule out rotator cuff tear in those with a history of disabling shoulder pain

Differential diagnosis

- Dislocated biceps tendon (tenderness over the bicipital groove but no distal muscle bulge)
- Distal biceps rupture
- Glenohumeral arthritis
- Impingement syndrome

Rotator cuff tear (often coexists with biceps tendon rupture)

Rupture of pectorals major

biceps tendon rupture treatment

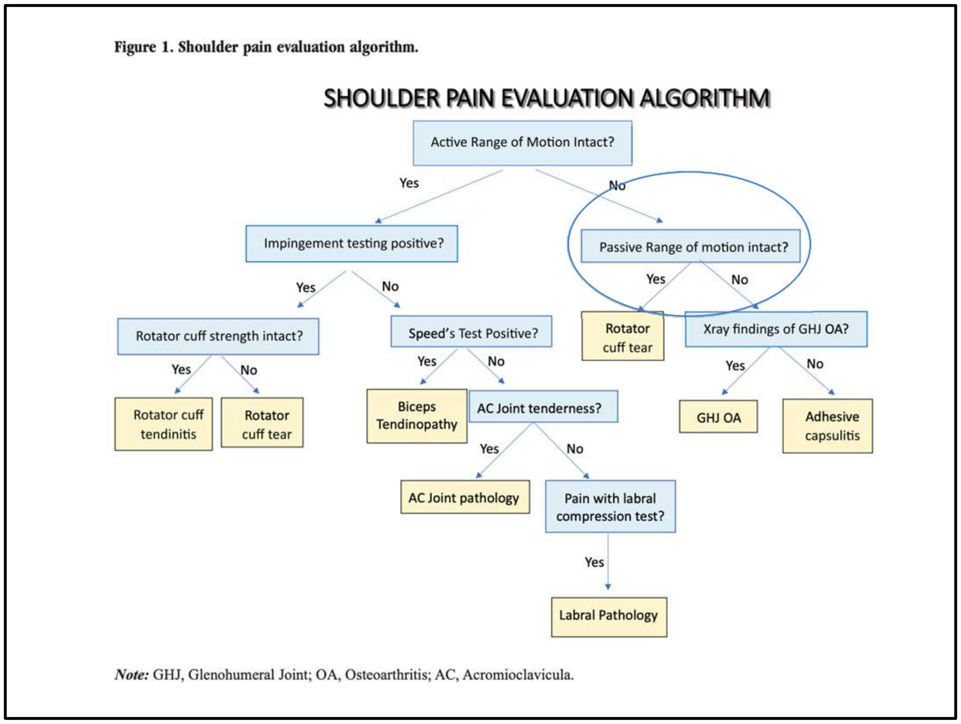
- Nonsurgical treatment is effective for most patients
  - Results in little loss of function and acceptable cosmetic deformity
  - Most patients regain full ROM and normal elbow flexion with physical therapy
- Young athletes and adults younger than 40 y.o. should be considered for surgical repair

Frieda
returns 2 months later…

- She admits that her pain and ROM seemed to improve after using NSAIDs. But then her pain became more intense and her shoulder is stiff. She can barely move it from her side.

- Examination:
- Inspection, no atrophy,
- Palpation, some tenderness anterior shoulder/deltoid. ROM: active ROM is limited flexion, abduction is about 60 degrees. She cannot move her R shoulder into external rotation. Passive ROM: similarly limited.

- What is this diagnosis?



Inspection, Palpation, Range of Motion, Strength

Adhesive Capsulitis (Frozen Shoulder)

- Idiopathic fibrosis of the joint capsule → loss of both active and passive ROM.
- Risk factors: Diabetes, thyroid disease, immobilization, female >40.
- History:
  - Severe, nagging pain at night
  - **Progressive** global stiffness of the shoulder over days to weeks
  - Significant disability of shoulder in ADLs, work, and leisure
  - Pain is usually around the deltoid but may radiate up

40-50% will have bilateral involvement.

Immobilization of the shoulder: other shoulder injuries can lead to the development of Adhesive capsulitis.

Stages of Adhesive Capsulitis

- Stages:
  - Freezing (pain) ->
  - Frozen (stiffness)->
  - Thawing (recovery)

Early "Freezing" phase of pain and progressive loss of motion

Pain with active and passive ROM

Pain at night and at rest

Thought to be inflammatory phase

Can last 3-6 months

"Frozen" stage

Patients no longer have pain with rest and at night

Significant shoulder stiffness

- **"Thawing" phase of decreasing discomfort associated with a slow but steady improvement in range of motion**
  - Can take 6 months to 2 years to resolve
  - Minimal long term pain or functional deficits

Adhesive Capsulitis: physical exam

- Significant (at least 50%) reduction in both active and passive range of motion
- Loss of external rotation with the arm at the side is pathognomonic
  - Loss of arm swing with gait
  - Motion limited and painful
- Pain at the deltoid insertion is common

due to contracture of the coracohumeral ligament

PHYSICAL EXAM:

Shoulder active range of motion in adhesive capsulitis. (A) Stiffness of shoulder abduction is shown in left side with the rotation of scapula, which means the loss of scapulohumeral rhythm. (B) Stiffness of shoulder external rotation in left side with normal range of motion of right side. (C) Stiffness of shoulder internal rotation in left side with normal range of motion of right side.

Lee, Ho Jun. (2014). Differential diagnosis of common shoulder pain. Journal of the Korean Medical Association. 57. 653. 10.5124/jkma.2014.57.8.653.

Adhesive Capsulitis Diagnosis

- Diagnosis is usually made clinically
- AP and axillary radiographs are indicated to evaluate for glenohumeral joint arthritis.
- Ultrasound can evaluate rotator cuff pathology and bursa

X ray to rule out OA, loose bodies, MRI findings will show a contracted capsule and loss of inferior pouch.

Ultrasound: Dynamic imaging of patients with frozen shoulder reveals limitations in the sliding movement of the supraspinatus tendon below the acromion during abduction [39]. Ultrasound is also helpful in ruling out pathology of the rotator cuff and bursa. (upto date)

Differential: Chronic posterior shoulder dislocation, Impingement syndrome, osteoarthritis, post traumatic shoulder stiffness, rotator cuff tear, tumor

Adhesive Capsulitis treatment

- Patient education: Recovery can take up 2-3 years!
- NSAIDs, nonnarcotic analgesics and moist heat followed by gentle stretching
  - Avoid narcotics
- TENS unit
- Physical therapy and home stretching (3-6 months atleast)
- Steroid injections
- Nonsurgical treatment is successful in 80-85% of patients

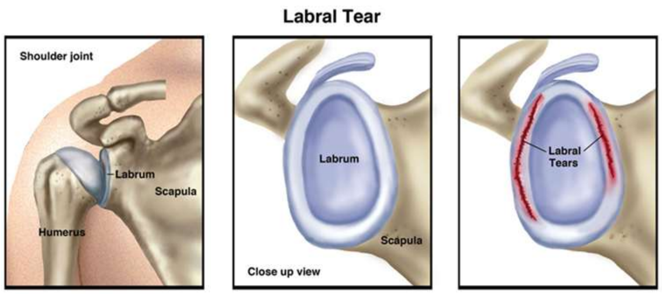
Operative: Ortho will take to OR and mechanically stretch under anesthesia. Caution with osteoporosis due to fracture risk.

- Referred for steroid injection at procedures clinic.
- Seen 1 month later with relief of pain but limited ROM.
- OMM, PT.
- Took 4 months to start to gain ROM again.

Glenohumeral Arthritis

Pain, limited ROM, X rays confirmatory,

PT, pain control, modification



Labral Injuries

SHOULDER AND UPPER ARM

Superior labrum anterior-to-posterior lesions (SLAP)

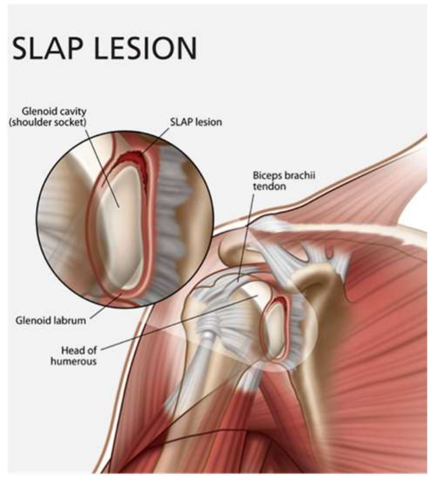
- SLAP lesions involve an injury to the superior glenoid labrum and the biceps anchor complex
  - Glenoid labrum provides increased stability of the shoulder
- Fraying or Tears: associated w repetitive overhead, fall on outstretched arm, traction on arm

DDx: AC joint arthritis
Biceps tendonitis
Rotator cuff injury
Shoulder instability
Subacromial impingement
Suprascapular nerve entrapment

slap clinical symptoms

- Symptoms:
  - Painful popping or mechanical catching in the shoulder
  - Vague deep shoulder pain
  - May report weakness or fatigue with use

  - Palpation: +/- tenderness
  - + O brien's Test



Difficult to diagnose and often a diagnosis of exclusion

After failure of PT and injections

slap diagnosis

- Plain radiographs are a good initial test but will not diagnose as SLAP lesion
- MR arthrography is the gold standard to evaluate for SLAP lesions
  - Sensitivity and specificity of up to 90%
  - Sensitivity of 50% for MRI without contrast
- Although imaging is important, definitive diagnosis can only be made through arthroscopy

slap treatment

- Nonsurgical
  - should be attempted first
  - NSAIDs, stretching and strengthening
  - In a throwing athlete, a gradual return to activities including an interval throwing program
- If nonsurgical care fails, shoulder arthroscopy may be needed

strain or spasm of accessory muscle

- Very common
- Often occurs after lifting or an injury
- May occur with cervical OA or torticollis
- Trapezius or Rhomboid spasm
- Treatment: Ice, heat, gentle stretching, OMM, hydration
  - Muscle relaxers, NSAIDs
  - Soft tissue injections sometimes effective

Stress, Ergonomics are triggers.
Other management is analyzing ergonomics at work/home, stress, and using other methods like OMM, stretching. Prevention in the future is to work on posture.

Exam is normal, hypertonic MM, address stress, ergonomics.

Key Takeaways

- Most shoulder pain arises from 4 core pathologies
- History + Physical + Ultrasound = 90% of your diagnosis.
- Start conservative unless red flags (acute tear, dislocation).
- Ultrasound enhances your Exam may help with dx or plan

Rotator Cuff Disease

GH arthritis

Adhesive Capsulitis

Shoulder Instability/ Labral d/s